Oracle SQL & PL/SQL

(40-Hour Instructor-Led Corporate Training)

Total Hours	Schedule	Session Length	Total Classes	Level
40 + 8–10 Bonus	**2 Days / Week**	**2.5 Hrs / Class**	**20 + Bonus**	**Beginner → Inter.**

Course Description

In this hands-on corporate training, participants will gain deep, practical skills in Oracle Database 19c — focusing on SQL query writing, PL/SQL programming, and enterprise development patterns. Through real-world banking and business scenarios, students will master how to retrieve, manipulate, and control data using Oracle SQL. They will advance into PL/SQL to write stored programs, packages, triggers, cursors, and bulk operations. A focused module on Oracle APEX demonstrates how PL/SQL powers real web applications. The course concludes with DBA fundamentals and a practical capstone project.

Training tools include Oracle Database 19c (or XE 21c), SQL Developer 23.x, Oracle LiveSQL, Oracle APEX, and Postman for REST API testing.

Who Should Attend

- Application developers and software engineers moving to Oracle environments
- Database administrators seeking strong SQL and PL/SQL programming foundations
- Business analysts and data professionals who need to query Oracle databases
- Fresh graduates and students targeting Oracle developer or DBA career paths
- IT professionals preparing for Oracle Certification (1Z0-071 or 1Z0-149)

Prerequisites

- Basic understanding of relational database concepts (tables, rows, columns) is helpful
- Familiarity with any programming language is an advantage (not mandatory)
- No prior Oracle or SQL experience required — this course starts from the very beginning

What You Will Accomplish

At the end of this course, students will be able to:

- Write SQL queries of all complexity levels using Oracle 19c syntax
- Use DML, DDL, DCL and TCL statements to manage and control database data
- Apply Oracle built-in functions: string, number, date, conversion, and NULL-handling
- Use all JOIN types, subqueries, SET operators, and analytic (window) functions
- Create and manage database objects: views, sequences, synonyms, indexes, and constraints
- Write complete PL/SQL programs using blocks, variables, control structures, and loops
- Build stored procedures, functions, and packages following industry standards
- Implement robust exception handling with predefined and user-defined exceptions
- Use explicit and implicit cursors, including the efficient cursor FOR LOOP
- Apply bulk operations using BULK COLLECT and FORALL for high-performance processing
- Write row-level and statement-level triggers for automation and auditing
- Use Dynamic SQL (EXECUTE IMMEDIATE) safely with bind variables
- Work with JSON data in Oracle 19c using JSON_VALUE, JSON_QUERY, and JSON_TABLE
- Understand Oracle APEX architecture and build a working web application on PL/SQL

- Apply industry naming conventions, package design standards, and security best practices
- Prepare confidently for Oracle SQL (1Z0-071) and PL/SQL (1Z0-149) certifications

What You Will Receive

- Complete digital course materials and all SQL/PL/SQL script files
- BankDB training schema — ready-to-use across all exercises
- Access to recorded class sessions for revision (if applicable)
- Assessment quiz sheets, assignment templates, and answer keys
- Final capstone project: pkg_banking — a complete Oracle package with APEX front-end
- Interview preparation guide: top SQL and PL/SQL questions with model answers
- Oracle certification roadmap and exam registration guidance

Follow-Up / Advanced Courses

- Oracle Database 19c Administration (DBA I & II)
- Oracle APEX Developer — Full Application Development
- Oracle Performance Tuning & Query Optimization
- Oracle REST Data Services (ORDS) & Web Services Development
- Oracle Data Warehousing & Analytics

CLASS-WISE TRAINING INDEX

Complete 20-Class Schedule + Bonus Classes

Class	Day	Topics Covered	Module	Duration
1	Week 1 – Day 1	Environment Setup: Oracle 19c/XE, SQL Developer, BankDB schema creation. Oracle Architecture overview. SELECT basics, column aliases, arithmetic, DUAL, DISTINCT	SQL Fundamentals	**2.5 hrs**
2	Week 1 – Day 2	WHERE clause (comparison, logical, BETWEEN, IN, LIKE, IS NULL). ORDER BY (ASC/DESC, multi-col, NULLS LAST). String functions: UPPER, LOWER, INITCAP, SUBSTR, INSTR, REPLACE, TRIM, LPAD	SQL Functions	**2.5 hrs**
3	Week 2 – Day 3	Number functions: ROUND, TRUNC, MOD, ABS, CEIL, FLOOR. Date functions: SYSDATE, ADD_MONTHS, MONTHS_BETWEEN, TO_DATE, TO_CHAR, LAST_DAY. Conversion & NVL/NVL2/COALESCE. CASE/DECODE	SQL Functions	**2.5 hrs**
4	Week 2 – Day 4	GROUP BY, HAVING. Aggregate functions: COUNT, SUM, AVG, MIN, MAX. WHERE vs HAVING distinction. Multi-level grouping. Business reporting queries on BankDB. SQL Quiz #1	SQL Aggregation	**2.5 hrs**
5	Week 3 – Day 5	JOINs: INNER JOIN, LEFT/RIGHT OUTER JOIN, FULL OUTER JOIN, CROSS JOIN. Self JOIN for hierarchical data (employee-manager). Multi-table JOINs (3+ tables). Venn diagram visual teaching	SQL JOINs	**2.5 hrs**
6	Week 3 – Day 6	Subqueries: single-row, multi-row (IN, ANY, ALL). Correlated subqueries. Inline views (subquery in FROM). ROWNUM vs FETCH FIRST (Oracle 12c+). EXISTS vs IN performance comparison	SQL Subqueries	**2.5 hrs**

Class	Day	Topics Covered	Module	Duration
7	Week 4 – Day 7	SET operators: UNION, UNION ALL, INTERSECT, MINUS. Analytic (Window) functions: RANK, DENSE_RANK, ROW_NUMBER, NTILE. Partitioned ranking. SQL Quiz #2	SQL Advanced	**2.5 hrs**
8	Week 4 – Day 8	Analytic functions continued: SUM/AVG OVER, LAG, LEAD, FIRST_VALUE, LAST_VALUE. Running totals. Month-over-month growth. CTEs (WITH clause). SQL Mini Project assigned	SQL Analytics + CTE	**2.5 hrs**
9	Week 5 – Day 9	DDL: CREATE, ALTER, DROP, TRUNCATE, RENAME, COMMENT. DML: INSERT (single/subquery), UPDATE (conditional), DELETE, MERGE (UPSERT). Constraints: PK, FK, UNIQUE, CHECK, NOT NULL	DDL / DML / Constraints	**2.5 hrs**
10	Week 5 – Day 10	TCL: COMMIT, ROLLBACK, SAVEPOINT. DCL: GRANT, REVOKE. Views (simple, complex, updatable, WITH READ ONLY). Sequences (NEXTVAL/CURRVAL). Synonyms. Indexes (B-tree, FBI). SQL Mini Project due	SQL Objects + TCL/DCL	**2.5 hrs**
11	Week 6 – Day 11	PL/SQL Architecture. Anonymous block structure (DECLARE/BEGIN/EXCEPTION/END). Variables: scalar, %TYPE, %ROWTYPE, CONSTANT, BOOLEAN. SELECT INTO. DBMS_OUTPUT. IF-THEN-ELSIF-ELSE, CASE	PL/SQL Core	**2.5 hrs**
12	Week 6 – Day 12	Loops: basic LOOP (EXIT WHEN), WHILE LOOP, FOR LOOP. Exception handling: predefined exceptions (NO_DATA_FOUND, TOO_MANY_ROWS, ZERO_DIVIDE, DUP_VAL_ON_INDEX, OTHERS). SQLCODE/SQLERRM	PL/SQL Control + Errors	**2.5 hrs**
13	Week 7 – Day 13	User-defined exceptions, RAISE, RAISE_APPLICATION_ERROR, PRAGMA EXCEPTION_INIT. Stored Procedures: IN/OUT/IN OUT params, SQL%ROWCOUNT. Error logging pattern. PL/SQL Quiz	Procedures + Exceptions	**2.5 hrs**
14	Week 7 – Day 14	Functions: syntax, RETURN, deterministic keyword. Procedure vs Function comparison. Package SPEC and BODY structure. Private vs public members. Package variables (session-level state)	Functions + Packages	**2.5 hrs**
15	Week 8 – Day 15	Cursors: implicit cursor attributes (SQL%FOUND, ROWCOUNT). Explicit cursor (OPEN/FETCH/CLOSE). Cursor FOR LOOP (preferred). Parameterized cursors. %ROWTYPE on cursors. Cursor exercises	Cursors	**2.5 hrs**
16	Week 8 – Day 16	Collections: Associative Arrays (INDEX BY TABLE), Nested Tables, VARRAYs. BULK COLLECT (full and chunked with LIMIT). FORALL for bulk DML. Performance comparison: row-by-row vs bulk	Collections + BULK	**2.5 hrs**
17	Week 9 – Day 17	Triggers: BEFORE/AFTER INSERT/UPDATE/DELETE (row & statement level), INSTEAD OF on views. :NEW/:OLD pseudorecords. Compound triggers. Audit trail trigger. Scheduler (DBMS_SCHEDULER)	Triggers + Scheduler	**2.5 hrs**
18	Week 9 – Day 18	Dynamic SQL: EXECUTE IMMEDIATE, bind variables, SQL injection prevention, DBMS_ASSERT. JSON in Oracle 19c: IS JSON constraint, JSON_VALUE, JSON_QUERY, JSON_TABLE, JSON_OBJECT. REST API with UTL_HTTP / APEX_WEB_SERVICE. pkg_banking capstone start	Dynamic SQL + JSON + REST	**2.5 hrs**

Class	Day	Topics Covered	Module	Duration
19	Week 10 – Day 19	Oracle APEX Essentials: Architecture (Browser→ORDS→DB), workspace, App Builder, Page Designer. Classic Report, Interactive Report, Interactive Grid. Page items, LOV, validations. PL/SQL Process (calling pkg_banking). Dynamic Actions. Authentication & Authorization basics	Oracle APEX	**2.5 hrs**
20	Week 10 – Day 20	DBA Basics: Users, Roles, Privileges, Tablespaces, Data Pump (expdp/impdp), Backup concepts, V$ views, Monitoring. Final Exam: SQL + PL/SQL combined (MCQ + coding). Course wrap-up. Career & certification roadmap	DBA Basics + Final Exam	**2.5 hrs**
B1	Bonus – Class 21	Advanced SQL: Recursive CTEs, PIVOT/UNPIVOT, LISTAGG, XMLAGG, MODEL clause intro, Pattern Matching (MATCH_RECOGNIZE intro), Invisible Indexes, Partitioned Tables overview	BONUS: Advanced SQL	**2 hrs**
B2	Bonus – Class 22	Advanced PL/SQL: Autonomous Transactions (PRAGMA AUTONOMOUS_TRANSACTION), UTL_FILE (file I/O), DBMS_CRYPTO (password hashing for OTP), Object Types, Pipeline Functions, DBMS_PIPE	BONUS: Adv. PL/SQL	**2 hrs**
B3	Bonus – Class 23	APEX Deep Dive: Calendar region, Charts & Dashboards (native charts), File Upload/Download, Email from APEX (APEX_MAIL), REST API consumption in APEX, Custom Authentication (OTP Login), CSS/JavaScript customization basics, Deployment & Application Export	BONUS: APEX Deep Dive	**2 hrs**
B4	Bonus – Class 24	Performance Tuning: EXPLAIN PLAN, DBMS_XPLAN, SQL Trace, Optimizer Hints, Index usage analysis, Bind variable peeking, AWR/ASH overview. Interview Mock: 20 timed SQL+PLSQL problems. Final Project Presentations	BONUS: Perf + Interview	**2 hrs**

Total Core Hours: 50 hrs | Bonus Hours: 8 hrs | Grand Total: ~58 hrs

Core: 20 classes x 2.5 hrs = 50 hrs | Bonus: 4 sessions x 2 hrs = 8 hrs | Assessment time included in class hours

COURSE OUTLINE: Oracle SQL & PL/SQL

PART I — SQL (Oracle Structured Query Language)

Write Basic SQL Queries

- Select data from one or more tables
- Apply column aliases and arithmetic expressions
- Use DISTINCT to eliminate duplicate rows
- Query the DUAL pseudo-table for expressions
- Order results using ORDER BY (ASC, DESC, NULLS LAST)

Apply Conditions with WHERE Clause

- Use comparison operators: =, !=, <, >, <=, >=
- Combine conditions: AND, OR, NOT
- Use BETWEEN, IN, NOT IN, LIKE (%, _ wildcards)
- Handle NULLs: IS NULL, IS NOT NULL

Use Oracle Single-Row Functions

- String: UPPER, LOWER, INITCAP, SUBSTR, LENGTH, INSTR, REPLACE, TRIM, LPAD, RPAD, CONCAT
- Number: ROUND, TRUNC, MOD, ABS, CEIL, FLOOR, POWER
- Date: SYSDATE, ADD_MONTHS, MONTHS_BETWEEN, TO_DATE, TO_CHAR, LAST_DAY, NEXT_DAY
- Conversion: TO_NUMBER, TO_DATE, TO_CHAR
- NULL handling: NVL, NVL2, NULLIF, COALESCE
- Conditional: CASE expression (simple and searched), DECODE

Group and Aggregate Data

- Use aggregate functions: COUNT, SUM, AVG, MIN, MAX
- Group data with GROUP BY (single and multi-column)
- Filter groups using HAVING clause
- Distinguish WHERE (row filter) vs HAVING (group filter)

Retrieve Data from Multiple Tables (JOINs)

- INNER JOIN — return only matching rows
- LEFT / RIGHT OUTER JOIN — preserve all rows from one side
- FULL OUTER JOIN — preserve all rows from both sides
- CROSS JOIN — Cartesian product for test data generation
- Self JOIN — query hierarchical/recursive data (employee-manager)
- Multi-table JOINs — 3 or more tables in a single query

Implement Advanced Queries

- Single-row and multi-row subqueries (IN, ANY, ALL)
- Correlated subqueries and the EXISTS operator
- Inline views (subquery in FROM clause)
- Scalar subqueries in SELECT list
- Top-N queries: ROWNUM and FETCH FIRST n ROWS ONLY

Use SET Operators

- UNION and UNION ALL — combine result sets
- INTERSECT — common rows between two queries
- MINUS — rows in first query not in second

Apply Analytic (Window) Functions

- RANK(), DENSE_RANK(), ROW_NUMBER() — ranking within partitions
- NTILE() — divide rows into equal buckets
- SUM/AVG/COUNT OVER — aggregates without collapsing rows
- Running totals using ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
- LAG() / LEAD() — compare current row to previous / next
- FIRST_VALUE / LAST_VALUE — boundary values in a window

Use DML to Manipulate Data

- INSERT: single-row and multi-row from subquery
- UPDATE: conditional updates using subqueries
- DELETE: targeted deletion with subquery filters
- MERGE (UPSERT): conditional insert or update in one statement
- SQL%ROWCOUNT, SQL%FOUND, SQL%NOTFOUND attributes

Use DDL to Define Schema Objects

- CREATE TABLE with all data types (NUMBER, VARCHAR2, DATE, CLOB)
- ALTER TABLE: add, modify, drop columns; rename tables
- DROP TABLE, TRUNCATE TABLE — differences and use cases
- RENAME and COMMENT on tables and columns

Work with Transaction Control (TCL)

- COMMIT — permanently save changes
- ROLLBACK — undo all uncommitted changes
- SAVEPOINT — create partial rollback markers
- Understand implicit commit (DDL) and session isolation

Control Database Access (DCL)

- GRANT object and system privileges to users and roles
- REVOKE privileges from users
- Create and manage named roles
- WITH GRANT OPTION for re-granting privileges

Create Database Objects

- Views: simple, complex, updatable, WITH READ ONLY, INSTEAD OF
- Sequences: NEXTVAL, CURRVAL, CYCLE, CACHE options
- Synonyms: private and public synonyms
- Indexes: B-tree, function-based (FBI), composite, invisible
- Use data dictionary: USER_TABLES, USER_CONSTRAINTS, USER_INDEXES

Define and Apply Constraints

- PRIMARY KEY, FOREIGN KEY (ON DELETE CASCADE / SET NULL)
- UNIQUE, CHECK, NOT NULL constraints
- Add, disable, enable, and drop constraints on existing tables
- Deferrable constraints for bulk data loads

Write Common Table Expressions (CTEs)

- WITH clause — named subquery for readability
- Multiple CTEs in one query (chained CTEs)
- Recursive CTEs for hierarchical data (org charts, BOM)

PART II — PL/SQL (Oracle Procedural Extension to SQL)

Understand PL/SQL Architecture

- PL/SQL engine vs SQL engine — context switching
- Anonymous blocks, named blocks (procedures, functions, packages, triggers)
- PL/SQL block structure: DECLARE / BEGIN / EXCEPTION / END
- Compilation, parsing, and execution model

Declare and Use Variables

- Scalar variables: VARCHAR2, NUMBER, DATE, BOOLEAN
- Anchored types: col%TYPE and table%ROWTYPE

Create Stored Procedures

- CREATE OR REPLACE PROCEDURE syntax
- IN, OUT, IN OUT parameter modes
- Default parameter values
- Calling procedures from SQL Developer, anonymous blocks, other procedures
- SQL%ROWCOUNT, SQL%FOUND, SQL%NOTFOUND in procedures

Write Stored Functions

- CREATE OR REPLACE FUNCTION with RETURN clause

- CONSTANT declarations — immutable values
- Custom RECORD types for structured data
- Variable scope and visibility rules
- DBMS_OUTPUT.PUT_LINE for debugging output

Apply Program Control Structures

- IF-THEN, IF-THEN-ELSE, IF-THEN-ELSIF-ELSE chains
- Simple and searched CASE statements
- LOOP with EXIT WHEN — basic iterative loop
- WHILE LOOP — pre-condition iteration
- FOR LOOP — numeric range iteration
- Nested loops with labeled EXIT

Handle PL/SQL Exceptions

- Predefined exceptions: NO_DATA_FOUND, TOO_MANY_ROWS, ZERO_DIVIDE, DUP_VAL_ON_INDEX, VALUE_ERROR, OTHERS
- SQLCODE and SQLERRM — capture error code and message
- DBMS_UTILITY.FORMAT_ERROR_BACKTRACE — full stack trace
- User-defined exceptions: EXCEPTION, RAISE
- RAISE_APPLICATION_ERROR — custom error codes (-20000 to -20999)
- PRAGMA EXCEPTION_INIT — bind exception to ORA error number
- Production error logging pattern: INSERT into error_log + RAISE

- DETERMINISTIC keyword for function-based index support
- Using functions inside SQL queries (SELECT, WHERE, ORDER BY)
- Procedure vs Function — when to use each

Design Packages (Industry Standard)

- Package SPECIFICATION (public interface)
- Package BODY (private implementation)
- Public vs private procedures, functions, variables, types
- Package-level (session-scoped) variables and constants
- Package initialization block (BEGIN...END in body)
- Overloading procedures and functions within a package
- Granting EXECUTE on a package
- Industry rule: all production code lives in packages

Use SQL*Plus and iSQL*Plus Tools

- SET SERVEROUTPUT ON SIZE UNLIMITED
- SHOW ERRORS — view compilation errors
- DESCRIBE — inspect object structure
- @ and @@ — run script files
- SPOOL — capture output to file
- Variable substitution: &var, &&var, DEFINE

Work with Cursors

- Implicit cursor — automatic for single-row DML and SELECT INTO
- Implicit cursor attributes: SQL%FOUND, SQL%NOTFOUND, SQL%ROWCOUNT, SQL%ISOPEN
- Explicit cursor — DECLARE, OPEN, FETCH, CLOSE lifecycle
- Cursor %ROWTYPE for structured fetch variable
- Cursor FOR LOOP — implicit open/fetch/close (preferred pattern)
- Parameterized cursors — pass values at OPEN time
- Cursor variables (REF CURSOR) — dynamic queries and SYS_REFCURSOR
- FOR UPDATE cursor with WHERE CURRENT OF for positioned DML

Apply Bulk Processing

- Why bulk: eliminate per-row context switching between PL/SQL and SQL engines
- BULK COLLECT — fetch entire result set into collection at once
- BULK COLLECT with LIMIT — chunked processing for memory efficiency
- FORALL — bulk DML (INSERT/UPDATE/DELETE) with collection data
- SAVE EXCEPTIONS — continue FORALL on partial failures
- SQL%BULK_ROWCOUNT — per-iteration DML count
- Performance benchmark: row-by-row vs BULK (10x–100x faster)

Write Triggers

Use Collections

- Associative Array (INDEX BY TABLE) — in-memory hash maps
- Nested Table — ordered collection storable in DB columns
- VARRAY — fixed-size ordered collection
- Collection methods: COUNT, EXISTS, FIRST, LAST, NEXT, PRIOR, EXTEND, DELETE, TRIM

- Row-level triggers (FOR EACH ROW): :NEW and :OLD pseudo-records
- Statement-level triggers (no FOR EACH ROW)
- BEFORE triggers — validate, set defaults, assign PKs
- AFTER triggers — audit log, cascade actions, notifications
- INSTEAD OF triggers — make complex/join views updatable
- Compound triggers — avoid mutating table errors
- ENABLE, DISABLE, DROP trigger management
- Restriction: no COMMIT/ROLLBACK inside row-level triggers

Apply Dynamic SQL

- EXECUTE IMMEDIATE — run SQL built at runtime
- Bind variables with USING clause — prevent SQL injection
- DBMS_ASSERT — validate object names in dynamic SQL
- Dynamic DDL — create/drop tables at runtime
- RETURNING INTO with EXECUTE IMMEDIATE
- OPEN cursor FOR dynamic query string

Work with JSON in Oracle 19c

- Store JSON in VARCHAR2, CLOB with IS JSON constraint
- JSON dot-notation access (Oracle 18c+)
- JSON_VALUE — extract scalar value from JSON
- JSON_QUERY — extract nested object or array
- JSON_TABLE — convert JSON arrays to relational rows
- JSON_OBJECT, JSON_ARRAY — build JSON from SQL queries
- JSON_EXISTS — conditional check on JSON path
- JSON indexes for query performance

Integrate REST APIs

- UTL_HTTP — call external REST endpoints from PL/SQL
- APEX_WEB_SERVICE.make_rest_request — simplified REST calls (when APEX installed)
- Parse JSON REST responses using JSON_VALUE / JSON_TABLE

Use Oracle Scheduler (DBMS_SCHEDULER)

- Create scheduled PL/SQL jobs (DBMS_SCHEDULER.create_job)
- Repeat intervals: FREQ=DAILY/WEEKLY/MONTHLY with BYHOUR/BYMINUTE
- Enable, disable, run, and drop jobs
- Monitor job status: USER_SCHEDULER_JOBS, USER_SCHEDULER_JOB_LOG
- Chain jobs for multi-step scheduled workflows

Apply Autonomous Transactions

- PRAGMA AUTONOMOUS_TRANSACTION — independent sub-transaction
- Use case: logging errors even when main transaction rolls back
- Commit and rollback independently inside autonomous block

Handle Files with UTL_FILE

- Open, read, and write OS files from PL/SQL
- Write reports and exports to server file system

- Send HTTP POST with JSON body from PL/SQL
- ACL (Access Control List) setup for network access

- UTL_FILE.FOPEN, PUT_LINE, GET_LINE, FCLOSE

Implement Security in PL/SQL

- Always use bind variables — never concatenate user input
- DBMS_CRYPTO for password hashing (SHA-256)
- Store OTP hashes instead of plain-text tokens
- DBMS_ASSERT.SQL_OBJECT_NAME for object name validation

PART III — Oracle APEX Essentials (Core: 1 Class | Bonus: 1 Class)

Understand APEX Architecture

- APEX components: Application Express engine, ORDS, Oracle DB
- Request lifecycle: Browser → ORDS → APEX Engine → DB → HTML
- Workspaces, schemas, and application containers
- APEX vs traditional web development — what APEX automates

Build Applications with App Builder

- Create new application: name, theme, authentication
- Page Designer: regions, items, buttons, attributes panel
- Classic Report region — write SQL, format columns, pagination
- Interactive Report — built-in filter, highlight, download, save
- Interactive Grid — inline editing, custom SQL, row actions
- Page items: text fields, select lists, date pickers, hidden items
- List of Values (LOV): static and dynamic LOV definitions
- Buttons: submit page, navigate, call JavaScript

Implement Page Logic

- Page Process: PL/SQL block executed on page submit
- Call pkg_banking procedures directly from APEX process
- Page computation: set item values before display
- Validations: required fields, PL/SQL custom validation

Use Dynamic Actions

- Event-based client-side actions without custom JavaScript
- Show/hide page regions based on item value
- Refresh a report region on item change (dynamic filtering)
- Execute PL/SQL block via Dynamic Action (AJAX call)
- Confirm before DML: dialog confirmation action
- Set item value dynamically from database query

Secure Your Application

- Authentication schemes: built-in database accounts, custom PL/SQL
- OTP (One-Time Password) custom login using DBMS_CRYPTO
- Authorization schemes: user role-based page/region access
- Session state: reading and setting item values in PL/SQL (:P1_ITEM)
- Session timeout and security settings

BONUS — Advanced APEX Topics

- Charts and Dashboards: native Oracle APEX chart types (Bar, Line, Pie, Gantt)
- File Upload: APEX_APPLICATION_FILES, move to permanent storage
- File Download: generate PDF/Excel from APEX or PL/SQL
- Email integration: APEX_MAIL.send from PL/SQL process
- REST API consumption: Web Source Modules in APEX
- Calendar and Timeline regions for scheduling applications

- Conditional display: show/hide regions and items by condition
- CSS customization: Universal Theme, custom CSS classes
- JavaScript integration: apex.item, apex.region, apex.event APIs
- Deployment: export/import application, publishing to production

PART IV — Oracle 19c DBA Fundamentals for Developers

Manage Users and Access

- Create and manage database users (CREATE USER, ALTER USER, DROP USER)
- Grant and revoke system privileges and object privileges
- Create, assign, and manage named roles
- WITH ADMIN OPTION and WITH GRANT OPTION
- Default and temporary tablespace assignment
- Quota management: UNLIMITED TABLESPACE, ALTER USER QUOTA

Understand Tablespaces and Storage

- Tablespace types: SYSTEM, SYSAUX, TEMP, UNDO, USER data
- View space usage: DBA_TABLESPACE_USAGE_METRICS
- Assign and query user quotas
- Data files and autoextend concept

Backup and Data Movement

- Data Pump Export (expdp): schema, table, and full database
- Data Pump Import (impdp): with REMAP_SCHEMA for environment clone
- Traditional Export/Import (exp/imp) — legacy awareness
- Concept of RMAN (Recovery Manager) for backups — overview only

Monitor and Maintain the Database

- Key V$ views: V$SESSION, V$SQL, V$PROCESS, V$LOCK, V$VERSION
- Data dictionary views: DBA_TABLES, DBA_INDEXES, DBA_CONSTRAINTS
- Identify blocking sessions and long-running queries
- AWR (Automatic Workload Repository) — overview for developers
- Alert log location and interpreting common ORA errors

PART V — Assessment, Projects & Industry Practices

Quizzes and Assessments

- SQL Quiz #1 (Class 4): SELECT, WHERE, ORDER BY, Functions — 15 MCQ + 3 queries
- SQL Quiz #2 (Class 7): JOINs, Subqueries, SET Operators — 10 MCQ + 4 queries
- PL/SQL Quiz (Class 13): Block structure, Exceptions, Procedures — 10 MCQ + 2 coding tasks
- SQL Mini Project (Class 10): 8 business reporting queries on BankDB — graded .sql file
- PL/SQL Package Project (Class 18): Complete pkg_banking — 10 components evaluated

Industry Best Practices

- Naming conventions: tables (PLURAL), views (vw_), procedures (verb_noun), packages (pkg_), triggers (trg_tablename_timing), variables (v_), parameters (p_), cursors (c_), constants (c_), types (t_)
- All production PL/SQL must live in packages — no standalone procedures
- Every block must have EXCEPTION ... WHEN OTHERS with logging + RAISE
- Always use %TYPE and %ROWTYPE — never hardcode data types
- Prefer cursor FOR LOOP over explicit OPEN/FETCH/CLOSE
- Use BULK COLLECT + FORALL for any loop processing 100+ rows

- APEX Application (Class 19): Working web app calling pkg_banking procedures
- Final Exam (Class 20): Combined SQL + PL/SQL — 15 MCQ + 4 coding tasks (90 min)

Practical Projects

- Banking Operations System (pkg_banking): open/close accounts, deposit, withdraw, transfer, bulk interest, audit triggers
- Student Management System: enrollment, grades, transcript view, reporting queries
- HR & Payroll System (pkg_payroll): payslip generation, increment processing, department reports
- Inventory & Stock System: stock in/out, low-stock alert trigger, reorder scheduler job
- Approval Workflow System (pkg_workflow): multi-step status machine, audit history
- OTP Login System: generate hash, validate, expire — DBMS_CRYPTO + APEX custom auth

- Never COMMIT or ROLLBACK inside a row-level trigger
- Always use bind variables in EXECUTE IMMEDIATE — prevent SQL injection
- Grant minimum required privileges — separate app user from DBA

Career and Certification Guidance

- Oracle Database SQL Certified Associate (1Z0-071) — after SQL module
- Oracle Database PL/SQL Certified Professional (1Z0-149) — after PL/SQL module
- Oracle APEX Developer Professional — after APEX module
- Oracle Database 19c Administrator Certified Professional (1Z0-082/083) — DBA path
- Career path: Junior Oracle Developer → Senior Developer → Technical Lead → Oracle Architect / DBA
- Top interview topics: JOINs, GROUP BY vs HAVING, analytic functions, package design, exception handling, BULK operations, trigger restrictions
- Recommended platforms: Oracle Dev Gym (devgym.oracle.com), Oracle LiveSQL, AskTom (asktom.oracle.com), oracle-base.com

Training Tools & Environment

Oracle DB 19c / XE 21c | SQL Developer 23.x | Oracle LiveSQL | Oracle APEX 23/24 | ORDS | Postman

Course prepared by an experienced Oracle Database Architect & Corporate Trainer | 2025–2026